Our Work

Assess performance -> Route requests -> Schedule execution -> Verify benefits

Shared evidence: benchmarks, request traces, resource assumptions and regional carbon signals

01 Performance

Task 1 / Evidence

Quality estimation

Pre-LightGBM
Response assessment

Resource accounting

Calling cost
Queue + power model

Evaluation outputs

Quality and cost
Latency and carbon

02 Routing

Task 2 / Model choice

Candidate comparison

Direct S / L
Sequential cascades

Decision safeguards

Quality-cost threshold
Task-specific risk guard

Selected policy

Direct Pre-LightGBM
Fallback to L

03 Scheduling

Task 3 / Region & time

Frozen model choices

Observed grid carbon
Capacity + deadlines

Spatial MILP + DES

Regional allocation
Queues and execution

Policy comparisons

Greedy
Temporal alternatives

04 Validation

Task 4 / Net impact

Stress scenarios

Task mix + quality noise
Demand growth

Service and fairness

All-input violations
Task-level quality / delay

Sustainability

Total facility carbon
Rebound + evidence

Decision support for SWITCH: adoption value, service trade-offs and environmental evidence

Historical replay and parameterized simulation; deployment savings require measurement.